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October 15, 1969
INTERIM REVISION OF
Fed. Spec. GGG-W-686b
February 5, 1963

INTERIM FEDERAL SPECIFICATION

WRENCH, TORQUE

This Interim Federal Specification was developed by the General Services Administration, Federal Supply Service, Standardization Division, Washington, D. C. 20406, based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above.

The General Services Administration has authorized the use of this Interim Federal Specification as a valid exception to Federal Specification GGG-W-686b, dated February 5, 1963.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers torque wrenches, also referred to as torque handles, used in applying a predetermined or measured torque to threaded fasteners or for other torque requirements.

1.1.1 Federal specification coverage. This specification does not cover all types, classes, and styles of the commodities indicated by the title of the specification or which are commercially available, but only the types, classes, and styles which are generally purchased by the Federal Government.

1.2 Classification.

1.2.1 Types. Torque wrenches shall be of the following types, classes, and styles, as specified (see 6.2):

Type I. Deflecting beam, direct reading, end drive.
- Class 1. Indicator plate, plain head.
  - Style A. Without signal, without memory indicator.
  - Style B. With (feel impulse) signal.
  - Style C. Without signal, with memory indicator.
- Class 2. Indicator dial, plain head.
  - Style A. Without signal, without memory indicator.
  - Style C. Without signal, with memory indicator.
- Class 3. Indicator plate, ratchet reversible head.
  - Style A. Without signal, without memory indicator.

Type II. Rigid case with indicator dial.
- Class 1. End drive.
  - Style A. Without signal, without memory indicator.
  - Style B. With visual signal, without memory indicator.
  - Style C. Without signal, with memory indicator.
- Class 2. Tee-handle (center drive).
  - Style A. Without signal, without memory indicator.
  - Style B. With visual signal, without memory indicator.
  - Style C. Without signal, with memory indicator.

Type III. Micrometer adjustable, presetting, (feel impulse).
- Class 1. Plain head (RH torquing).
- Class 2. Ratchet reversible head (RH torquing).
- Class 3. Plain head (RH and LH torquing).
- Class 4. Ratchet reversible head (RH and LH torquing).

Type IV. Tee-handle, ratcheting, preset.

FSC 5120

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2. APPLICABLE DOCUMENTS

2.1 Specifications and standards. The following specifications and standards, of the issues in effect on date of invitation for bids, or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

QQ-C-320 - Chromium Plating (Electrodeposited).
QQ-P-416 - Plating, Cadmium (Electrodeposited).
QQ-Z-325 - Zinc Coating, Electrodeposited, Requirements For.
TT-E-489 - Enamel Alkyd, Gloss (for Exterior and Interior Surfaces).
TT-L-58 - Lacquer; Spraying, Clear and Pigmented (General Use).
GGG-W-641 - Wrench, Bolt and Nut; Socket, Hand (Including Handles and Attachments).

Federal Standards:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civilian Agencies).
Fed. Std. No. 151 - Metals; Test Methods.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, D.C., Atlanta, Chicago, Kansas City, Mo., Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, Washington.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specifications:

MIL-A-8625 - Anodic Coatings, For Aluminum and Aluminum Alloys.
MIL-C-13924 - Coating, Oxide, Black, For Ferrous Metals.
MIL-H-15424 - Hand Tools, Packaging Of.
MIL-P-16232 - Phosphate Coatings, Heavy, Manganese or Zinc Base (For Ferrous Metals).

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issues in effect on date of invitation for bids, or request for proposal shall apply:

American Society for Testing and Materials (ASTM) Standard:

E29-67 - Indicating Significant Places in Limiting Values.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa. 19103.)

3. REQUIREMENTS

3.1 Illustrations. The illustrations herein are descriptive and not restrictive and are not intended to preclude the purchase of torque wrenches otherwise conforming to this specification.

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3.2 Material. The material to be used in the construction of torque wrenches shall be of uniform quality and construction, and shall conform in composition, heat treatment, and suitability to the standard practices of manufacturers producing equipment of the type called for in this specification. Where handgrips or other parts of the wrenches are made of plastic, they shall be shatterproof and nonflammable (see 4.4.5 thru 4.4.5.2).

3.3 Working parts. Working parts shall be suitably designed and processed to withstand ordinary service without showing excessive wear.

3.4 Measuring elements. Measuring elements used in the construction of the wrenches shall be of high quality to give extended service and repeat accuracy. The measuring mechanism of rigid case and indicator dial deflecting beam styles shall be protected to the extent practicable to exclude the entrance of foreign matter that would affect the accuracy, serviceability or durability of the wrench.

3.5 Metal frame cases, covers, heads, handgrips, and other parts. The metal frame cases, covers, heads, handgrips, and other parts shall be capable of withstanding the torque stresses imposed in the tests of 4.4.1 and 4.5 without permanent set or breakage. When handgrips are of hollow construction as in pivoted type handgrips, the walls shall be of sufficient thickness and strength so as not to take permanent set, crack, or break during normal use or on being subjected to the test of 4.4.1.

3.6 Location of torque presetting device. The torque presetting device on type I and II wrenches shall be located in the vicinity of the indicator dial or plate or in any practical area provided it performs satisfactorily. It shall be accurate and shall incorporate features to prevent the setting from changing while the wrench is in use. Presetting the torque value shall be accomplished without the use of tools or other equipment.

3.7 Drive requirements. The torque wrenches shall be provided with a square drive in accordance with GGG-W-641, except ball and spring, and hardness shall be Rockwell C 34 to 52. The opposite flats of the square drive shall be parallel within 5° of the longitudinal axis of the handle of plain head wrenches. Ball detent shall be located as shown in figure 1.

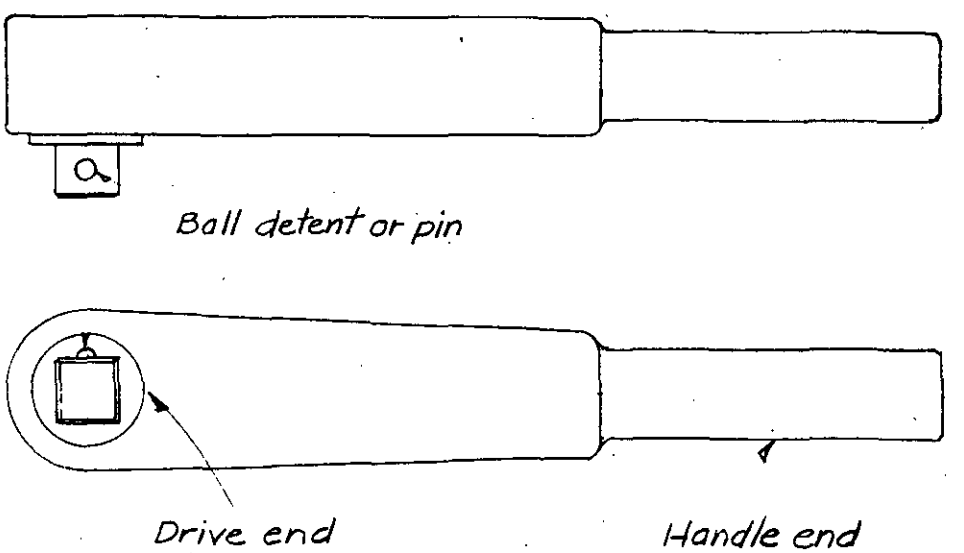


FIGURE 1. Location of ball detent or pin on fixed head torque wrenches.

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3.8 Position of indicator dials or scales. Indicator dials or scales shall be integral or securely attached to the wrench and shall indicate the torque being applied on either side of zero to the types I and II wrenches. The indicator dial or scales shall be so positioned on the wrench as to provide ease of reading when the wrench is in use.

3.8.1 Indicator markings. Each indicator, dial, plate, and barrel shall be marked "inch-grams", "inch-ounces", "inch-pounds", or "foot-pounds" as appropriate. The torque scale from minimum to the maximum value shall be graduated as indicated for the various types and numbered as appropriate. The scale graduations and identifying figures shall be permanent, legible, and of adequate size for easy reading in both daylight and artificial light. The face of each dial and scale shall be of sufficient size to permit adequate spacing of the graduations. This paragraph does not apply to type IV. The width of any graduation line shall be less than the adjacent space width.

3.8.2 Dial crystal. The crystal of the dials shall be of a type with high resistance to accidental breakage and shall remain clear and free from deterioration. The surface of the crystal shall be protected to allow the torque wrench to be dropped or slid over a workbench without scratching or breaking the crystal. This protection may be a part of the housing, or the bezel may be so designed as to allow sufficient clearance between its top face and the top surface of the crystal. The protective means shall in no way obstruct or cover any part of the dial face.

3.8.3 Pointer clearance (parallax). Where the pointer of an indicator plate or of a dial is not provided with means for avoiding errors in reading due to the parallax, other than sighting over or down the length of the pointer. the clearance between the pointer index and the graduated scale shall not exceed 0.030 inch or the width of spacing between increments whichever is greater,and shall not touch the scale at any position of the pointer.

3.8.4 Width of indicating point. In wrenches having indicator plates or dials, the maximum tip or line width of the indicating point or pointer shall not exceed the narrowest graduation line width plus 25 percent, on the indicator plate or dial. Pointers tapering in width with a radius or flat on the end shall have a diameter or width of flat not to exceed the graduation line width plus 25 percent. The pointer shall not cover the entire length of the increment line.

3.9 Signal mechanisms. All signaling style wrenches shall be equipped with one of the signal devices specified in 3.9.1 or 3.9.2. They shall be accurate and shall incorporate features to prevent the setting from changing while the wrench is in use. Setting the torque indicating device shall be accomplished without the use of tools or other equipment.

3.9.1 Feel impulse. Wrenches equipped with feel impulse shall contain a torque limiting device which shall momentarily release automatically and signal when the predetermined torque has been reached by feel impulse with or without audible signal.

3.9.2 Visual signal. Wrenches equipped with a visual signal shall have the visual signaling device, located in the vicinity of the indicator plate or dial, which shall flash a light when the preselected torque has been applied. The indicating light shall incorporate a cover which shall extend over and protect the bulb. The light shall be powered by a standard dry cell battery. Each wrench shall include a bulb in good operating condition when shipped.

3.10 Memory indicator. Wrenches equipped with memory indicator shall have an auxiliary pointer capable of registering the maximum applied torque. The pointer shall be so designed as to retain the reading during normal handling after loading. Provision shall be made to reset the indicator as required. The design of the pointer shall conform to paragraph 3.8.4. The pointer shall be capable of withstanding the overload test of 4.4.1.

3.11 Maximum capacity type I, II, and III wrenches. Wrenches of a greater maximum capacity meeting the other physical requirements of the specific sizes will be acceptable and shall be tested up to the maximum capacity. Wrenches shall be of such length that a maximum force of 175 pounds when applied to the center of the handgrip or to an extension adapter shall be sufficient to apply the maximum rated torque without changing the specified accuracy.

3.12 Accuracy and direction.

3.12.1 Unless otherwise specified, the wrenches shall indicate, in a right-hand and left-hand direction, the applied torque to the tolerances specified in table I. The pointers of the types I and II wrenches shall return to zero after use. When specified (see 6.2), types I and II wrenches shall be accurate within 2 percent of reading from 20 percent to 100 percent of capacity. When specified (see 6.2), types I and II wrenches shall be accurate within 1 percent of reading from 20 percent to 100 percent of capacity. All specified tolerance limits are absolute limits, as specified in ASTM recommended practice E29-67.

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TABLE I. Wrench tolerances

Percent of full scale or capacity		Tolerance plus or minus percent of reading			
From	Including	Types I and II	Type III		Type IV
			Clockwise	Counterclockwise	
0	19	10 percent	---	---	Tolerance = 5% of setting, or, as specified below whichever is greater:
20	Full scale	4 percent	---	---	Size 1: ± 1/2 inch-ounce. Size 2: ± 1/2 inch-pound.
Minimum setting	Full scale	---	4 percent	6 percent	

3.12.2 Recalibration. Types II, III, and IV torque wrenches shall be capable of being recalibrated to the accuracy requirements of table I, whenever calibration is lost due to normal use not involving mishandling.

3.13 Finish and coatings. All external wrench parts shall be smoothly finished. External parts, except those inherently corrosion-resistant, or the ball and spring on square drive, or of polished aluminum, shall be protected with one or a combination of any of the following coatings at the option of the manufacturer:

(a) Chemically produced oxide coating, oil treated.
(b) Chemically produced phosphate coating, oil treated.
(c) Chrome, cadmium, zinc, or equally suitable plating.
(d) Lacquered or enameled.
(e) Anodic coating for aluminum.

3.13.1 Coatings. The coatings shall be adherent, smooth, continuous, and free from pits, blisters, nodules, and any other defects which would interfere with their protective value and serviceability.

3.13.1.1 Chromium plate. The coating shall be electrodeposited metals consisting of nickel or nickel and copper followed by chrome, the minimum thickness being 0.0002 inch for nickel or nickel and copper, and 0.00001 inch for chrome. Plating adherence and thickness shall be tested in accordance with the test requirements for class I plating of QQ-C-320.

3.13.1.2 Cadmium plate. The coating shall be in accordance with QQ-P-416 class 2, type II.

3.13.1.3 Zinc plate. The coating shall be in accordance with QQ-Z-325 type I, class 2.

3.13.1.4 Hot phosphating. The coating shall be in accordance with MIL-P-16232 type M, class 2 or type Z, class 2.

3.13.1.5 Oxide coating. The coating shall be in accordance with MIL-C-13924 class 1 with a coating of preservative oil.

3.13.1.6 Enamel. The enamel coating shall be in accordance with TT-E-489 or other synthetic enamels of equivalent quality.

3.13.1.7 Lacquer. The lacquer coating shall be in accordance with TT-L-58 or other synthetic lacquers of equivalent quality.

3.13.1.8 Anodic coating. The coating shall be in accordance with MIL-A-8625.

3.13.2 Coating internal parts. Internal parts subject to corrosion shall be protected by the use of oil or other protective coating.

3.14 Marking. Each wrench shall be marked in a legible and permanent manner with the model number, and manufacturer's name or with a trademark of such known character that the source of manufacture may be readily determined. Type III, classes 3 and 4, wrenches shall be permanently marked "indicates right and left".

3.15 Carrying case. Unless otherwise specified (see 6.2), a case shall be supplied for each type I, II, and III wrench. The case shall be sufficiently heavy-gage sheet steel that will not distort due to the weight of the wrench, and shall be of the following minimum gages: No. 24 for wrenches weighing under 5 pounds, No. 22 for wrenches weighing 5 to 15 pounds, and No. 20 for wrenches weighing over 15 pounds. Plastic or other sheet metal of equivalent performance characteristics

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may be employed. Permanently fixed means shall be made in the case to hold the wrench securely in position to protect it from damage. The case shall also be provided with a hinged cover with suitable catch, and if of steel, shall have a durable coating of paint or enamel. The color of the paint or enamel, if other than the manufacturer's standard, shall be as specified by the procuring activity. The case shall be marked as follows:

Wrenches, Torque Capacity _____.
Manufacturer's name or trademark.

3.16 Type I, deflecting beam. Type I wrenches shall operate by the flexing of a cantilever beam or beams. The beams may be partially enclosed. The drive square or tang shall be located at one end of the beam and the handgrip with the indicator plate or dial at the other. The handle design must provide a means for fixing the lever length of the torque wrench to provide an accurate means for the correct use of torque wrench adapter combinations.

3.16.1 Deflecting element. All exposed surfaces of the deflecting element shall be free of markings and stamp and tool marks.

3.16.2 Class 1, indicator plate, plain head. Class 1 wrenches shall have a dial plate and shall have either one or two pointers for indicating the torque being applied on the wrench. The pointer or pointers shall be integral with the head and drive square. Wrenches shall conform to table II for the size specified (see 6.2). Wrenches of a torque capacity or drive size other than those listed in table II shall be, as specified (see 6.2).

TABLE II. Type I, class 1, deflecting beam, indicator plate, (direct reading, plain head)

Torque capacity				Square drive (inch)	Increment graduations (max.)	Style		
Inch-grams	Inch-ounces	Inch-pounds	Foot-pounds			A	B	C
0-80	----	----	----	1/4	5 in. - grams	x		
----	0-32	----	----	1/4	2 in. - ounces	x		
----	0-80	----	----	1/4	5 in. - ounces	x		
----	----	0-10	----	1/4	1/2 in. - pounds	x		x
----	----	0-25	----	1/4	1 in. - pounds	x		x
----	----	0-60	----	3/8	2 in. - pounds	x	x	x
----	----	0-120	----	3/8	5 in. - pounds	x	x	x
----	----	0-200	----	3/8	10 in. - pounds	x	x	x
----	----	0-300	----	3/8	10 in. - pounds	x	x	x
----	----	0-600	----	3/8	25 in. - pounds	x		
----	----	----	0-50	3/8	2 ft. - pounds	x	x	x
----	----	----	0-150	1/2	5 ft. - pounds	x	x	x
----	----	----	0-300	3/4	10 ft. - pounds	x	x	x
			0-600	3/4	10 ft. - pounds	x	x	x

x = Available in this style.

3.16.2.1 Style A, direct reading. Type I, class 1, style A wrenches shall indicate torque visually on an indicator plate. Wrenches shall be similar to figure 2.



FIGURE 2. Type I, class 1, style A, torque wrench, defecting beam, with indicator plate, (direct reading).

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3.16.2.2 Style B, direct reading with feel impulse signal. Style B torque wrenches shall meet the requirements of the type I, class 1, style A wrenches and in addition shall have a feel impulse signaling device in accordance with 3.9.1. Wrenches shall be similar to figure 3.

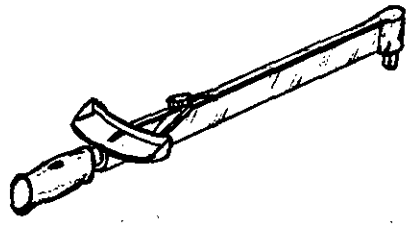


FIGURE 3. Type I, class 1, style B, torque wrench, deflecting beam, with indicator plate, (direct reading with feel impulse signal).

3.16.2.3 Style C, direct reading with memory indicator. Style C torque wrenches shall meet the requirements of the type I, class 1, style A wrenches and in addition shall have a memory indicator in accordance with 3.10. Wrenches shall be similar to figure 4.

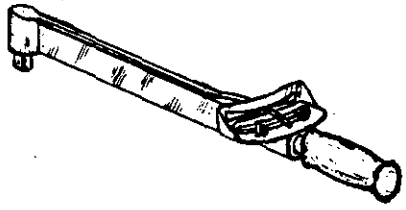


FIGURE 4. Type I, class 1, style C, torque wrench, deflecting beam, with indicator plate, (direct reading with memory indicator).

3.16.3 Class 2, indicator dial, plain head. Wrenches shall conform to table III for the size specified (see 6.2). Wrenches of a torque capacity or drive size other than these listed in table III shall be, as specified (see 6.2). If specified (see 6.2), the 0-600 foot-pound size shall be the center drive style with fixed handles.

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TABLE III. Type I, class 2, deflecting beam, indicator dial, (direct reading, plain head)

Torque capacity		Square drive (inch)	Increment graduations (max.)	Style	
Inch-pounds	Foot-pounds			A	C
0-100	----------	1/4	5 in. - pounds	x	x
0-250	----------	1/4 or 3/8	10 in. - pounds	x	x
-----	0-50	3/8	2 ft. - pounds	x	x
-----	0-150	1/2	5 ft. - pounds	x	x
-----	0-300	3/4	10 ft. - pounds	x	x
-----	0-600	3/4	10 ft. - pounds	x	x
-----	0-1000	1	50 ft. - pounds	x	x
-----	0-2000	1	50 ft. - pounds	x	x

x = Available in this style.

3.16.3.1 Style A, direct reading. Type I, class 2, style A wrenches shall indicate torque visually on an indicator dial. Wrenches shall be similar to figure 5.

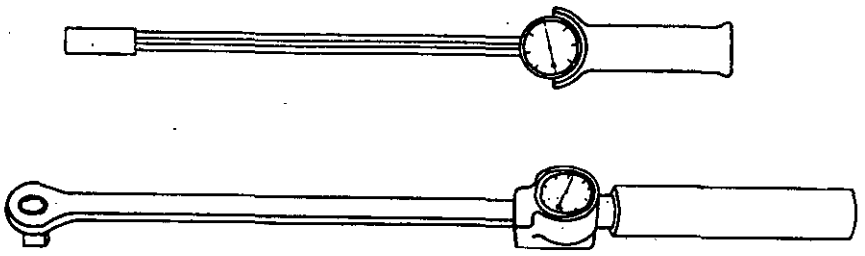


FIGURE 5. Type I, class 2, style A, torque wrenches, deflecting beam, with indicating dial, (direct reading).

3.16.3.2 Style C, direct reading. Type I, class 2, style C wrenches shall meet the requirements of the type I, class 2, style A wrenches and in addition shall have a memory indicator in accordance with 3.10. Wrenches shall be similar to figure 6.

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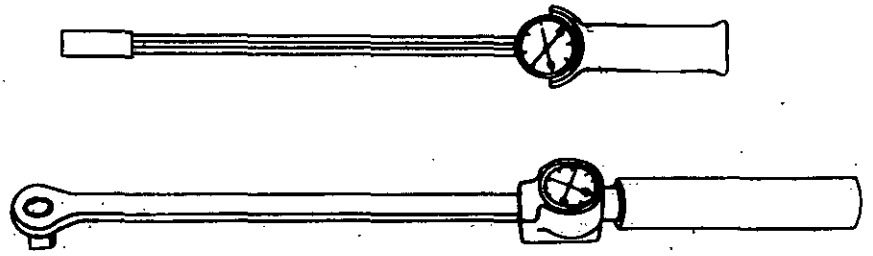


FIGURE 6. Type I, class 2, style C, torque wrenches, deflecting beam, with indicating dial, (direct reading, with memory indicator).

3.16.4 Class 3, indicator plate, ratchet reversible head. Class 3 wrenches shall conform to the requirements for type I, class 1, wrenches, except that they shall contain a reversible ratchet mechanism enclosed in the drive head. The ratcheting device shall meet the requirements of GGG-W-641. The reversing device, pawl and ratchet wheel shall be of steel suitably hardened for hard usage. Wrenches shall conform to table IV for the size specified (see 6.2). Wrenches of a torque capacity or drive size, other than those listed in table IV, shall be as specified (see 6.2).

TABLE IV. Type I, class 3, style A, deflecting beam, indicator plate, (direct reading, ratchet reversible head)

Torque capacity		Square drive	Increment graduations
Inch-pounds	Foot-pounds	(Inch)	(max.)
0-600	---	3/8 or 1/2	25 in.-pounds
---	0-150	1/2	5 ft.-pounds

3.16.4.1 Style A, direct reading. Type I, class 3, style A wrenches shall indicate torque visually on an indicator plate. Wrenches shall be similar to figure 7.

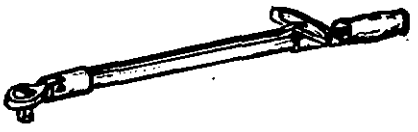


FIGURE 7. Type I, class 3, style A, torque wrench, defelecting beam, indicator plate, (direct reading, with ratchet reversible head).

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3.17 Type II, rigid case, with indicator dial. Type II wrenches shall have actuating mechanism enclosed in a stiff or rigid frame. A removable cover shall be provided for the purpose of inspection, calibration and repair. A dial indicator shall be provided for visually indicating both the right-hand and left-hand torque being applied.

3.17.1 Class 1, end drive. Class 1 wrenches shall have the square or tang located at one end of the wrench and the handgrip with indicator dial at the other. Wrenches shall conform to table V for the size specified (see 6.2). Wrenches of a torque capacity, or drive size other than those listed in table V shall be as specified (see 6.2).

TABLE V. Type II, class 1, rigid case, with indicator dial, (direct reading, end drive)

Torque capacity			Square drive	Increment graduations (max.)	Style		
					A	B	C
Inch-ounces	Inch-pounds	Foot-pounds	Inch				
0-240	----------	----------	1/4	10 in. - ounces	x	x	x
-----	0-30	----------	1/4	1 in. - pounds	x	x	x
-----	0-75	----------	1/4	1 in. - pounds	x	x	x
-----	0-150	----------	3/8	5 in. - pounds	x	x	x
-----	0-300	----------	3/8	5 in. - pounds	x	x	x
-----	0-600	----------	3/8	20 in. - pounds	x	x	x
-----	0-1200	----------	1/2	20 in. - pounds	x	x	x
-----	0-1800	----------	1/2	60 in. - pounds	x	x	x
-----	0-4200	----------	3/4	120 in. - pounds	x	x	x
-----	0-7200	----------	3/4	120 in. - pounds	x	x	x
	----------	0-50	3/8	2 ft. - pounds	x	x	x
	----------	0-175	1/2	5 ft. - pounds	x	x	x
	----------	0-250	3/4	10 ft. - pounds	x	x	x
	----------	0-350	3/4	10 ft. - pounds	x	x	--
	----------	0-420	3/4	10 ft. - pounds	x	x	x
	----------	0-600	3/4	10 ft. - pounds	x	x	x
	----------	0-1000	1	25 ft. - pounds	x	x	x
	----------	0-2000	1-1/2	50 ft. - pounds		x	x

x = Available in this style.

3.17.1.1 Style A, direct reading. Type II, class 1, style A wrenches shall indicate torque visually on an indicator dial. Wrenches shall be similar to figure 8.

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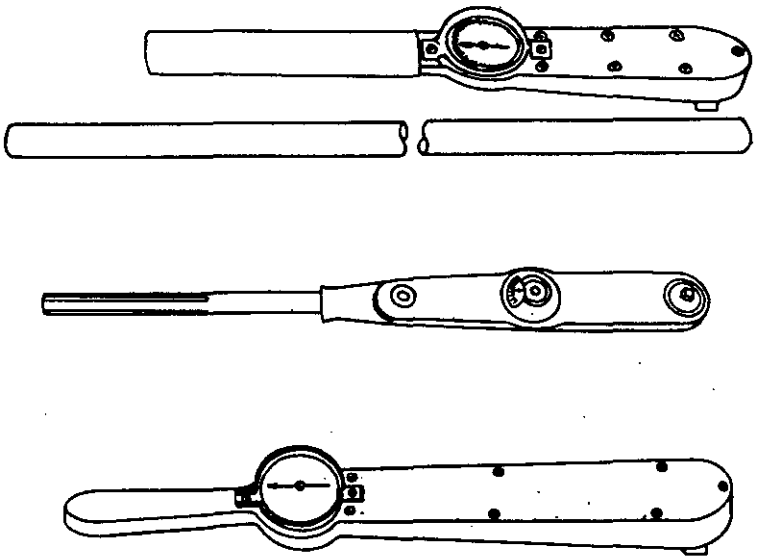


FIGURE 8. Type II, class 1, style A, torque wrenches, rigid case, with indicator dial, (direct reading).

3.17.1.2 Type II, class 1, style B, direct reading and visual signal. Style B wrenches shall meet the requirements of type II, class 1, style A wrenches and in addition shall have a visual signal in accordance with 3.9.2. Wrenches shall be similar to figure 9.

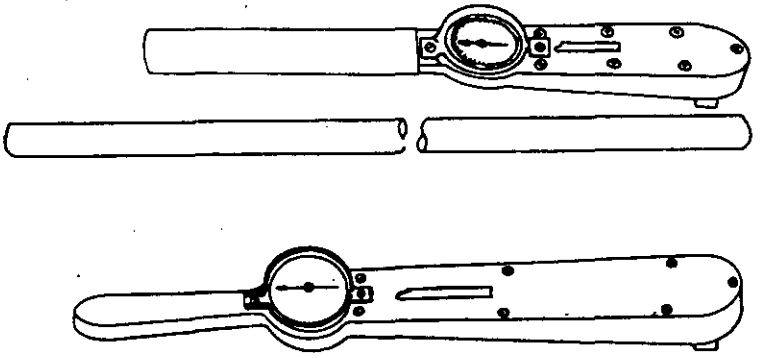


FIGURE 9. Type II, class 1, style B, torque wrench, rigid case, with presetting dial, (direct reading, with visual signal).

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3.17.1.3 Type II, class 1, style C, direct reading, with memory indicator. Style C wrenches shall meet the requirements of type II, class 1, style A wrenches and in addition shall have a memory indicator in accordance with 3.10. Wrenches shall be similar to figure 10.

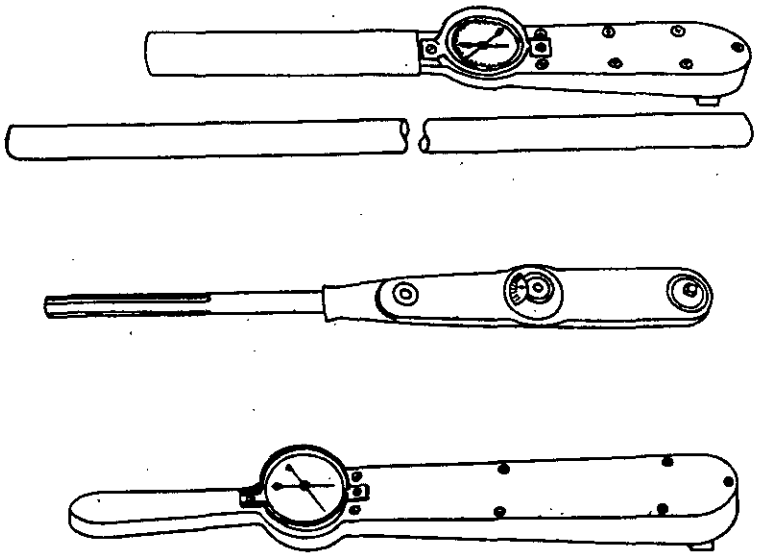


FIGURE 10. Type II, class 1, style C, torque wrenches, rigid case, with indicator dial,(direct reading, with memory indicator).

3.17.2 Class 2, tee-handled, (center drive). Type II, class 2 wrenches shall be the tee-handled style with the dial indicator and drive square at the center.

3.17.2.1 Adapter handles. Wrenches 1500 foot-pound capacity and larger shall have two tubular slip fit adapter handles with holes for engagement with the spring back plungers provided in the tee-handle of the wrench.

3.17.2.2 Two-man operation. Torque power may be applied from both sides of the square drive simultaneously or the total handle length can be moved to one side for one-man operation.

3.17.2.3 Wrench sizes. Wrenches shall conform to table VI for the size specified (see 6.2). Wrenches of a torque capacity or drive size, other than those listed in table VI, shall be as specified (see 6.2).

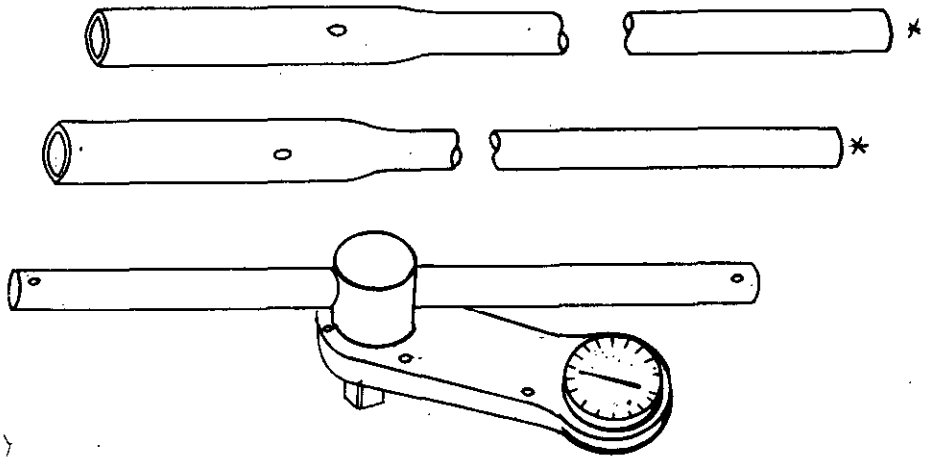
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TABLE VI. Type II, class 2, rigid case, with indicator dial, (direct reading, tee-handle, center drive)

Torque capacity			Square drive	Increment graduations (max.)	Style		
					A	B	C
Inch-ounces	Inch-pounds	Foot-pounds	Inch				
0-240	---	---	1/4	10 in.-ounces	x		x
---	0-30	---	1/4	1 in.-pounds	x		x
---	0-75	---	1/4	1 in.-pounds	x		x
---	0-150	---	3/8	5 in.-pounds	x		x
---	0-300	---	3/8	5 in.-pounds	x		x
---	0-600	---	3/8	20 in.-pounds	x		x
---	0-1200	---	1/2	20 in.-pounds	x		x
---	0-1800	---	1/2	60 in.-pounds	x		x
---	0-4200	---	3/4	120 in.-pounds	x	x	x
---	0-7200	---	3/4	120 in.-pounds	x	x	x
	---	0-50	3/8	2 ft.-pounds	x		x
	---	0-175	1/2	5 ft.-pounds	x		x
	---	0-250	3/4	10 ft.-pounds	x		x
	---	0-350	3/4	10 ft.-pounds	x	x	x
	---	0-420	3/4	10 ft.-pounds	x	x	x
	---	0-600	3/4	10 ft.-pounds	x	x	x
	---	0-1000	1	25 ft.-pounds	x	x	x
		0-2000	1-1/2	50 ft.-pounds		x	x

x = Available in this style.

3.17.2.4 Style A, direct reading. Type II, class 2, style A wrenches shall indicate torque visually on an indicator dial. Wrenches shall be similar to figure 11.

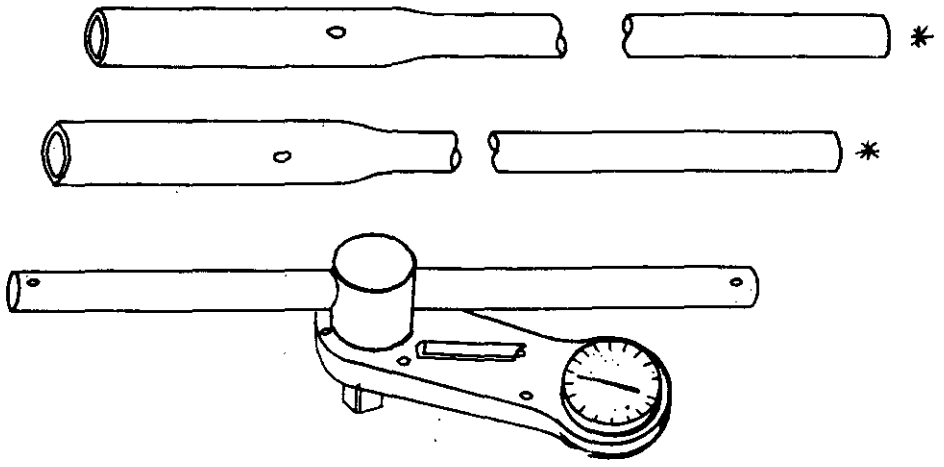


*Required for wrenches 1500 foot-pound capacity or greater.

FIGURE 11. Type II, class 2, style A, torque wrench, rigid case, with indicator dial, (direct reading).

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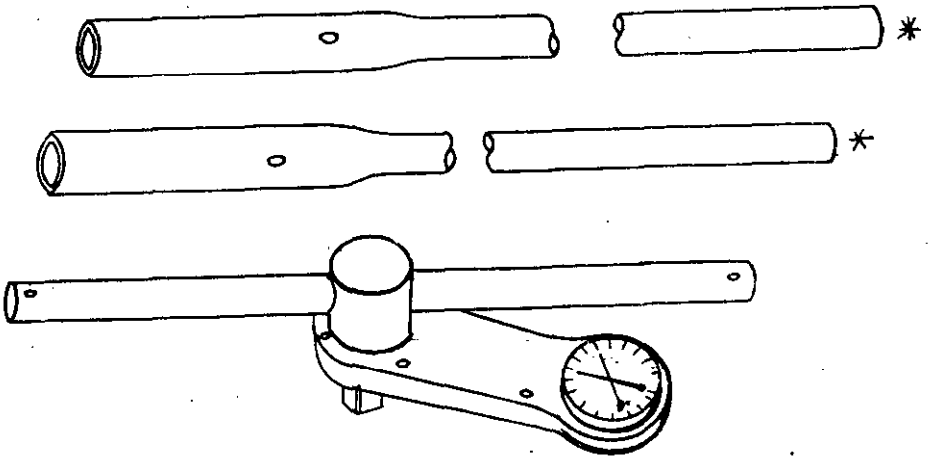
3.17.2.5 Type II, class 2, style B, direct reading and visual signal. Style B wrenches shall meet the requirements of type II, class 2, style A, wrenches and in addition shall have a visual signal in accordance with 3.9.2. Wrenches shall be similar to figure 12.



* Required for wrenches 1500 foot-pound capacity or greater.

FIGURE 12. Type II, class 2, style B, torque wrench, rigid case, with presetting dial, (direct reading, with visual signal).

3.17.2.6 Type II, class 2, style C, direct reading, with memory indicator. Style C wrenches shall meet the requirement of type II, class 2, style A wrenches and in addition shall have a memory indicator in accordance with 3.10. Wrenches shall be similar to figure 13.



*Required for wrenches 1500 foot-pound capacity or greater.

FIGURE 13. Type II, class 2, style C, torque wrench, rigid case, with indicator dial, (direct reading, with memory indicator).

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3.18 Type III, micrometer adjustable, torque presetting. Type III wrenches shall be provided with an accurate micrometer-type setting upon which the desired torque can be set by turning the adjustable handle. When the torque applied equals the setting on the handle, the wrench shall momentarily release and signal that the predetermined torque has been reached by feel impulse with or without audible signal. When hand pressure is removed from handle, the wrench shall automatically reset for the next operation. Means shall be provided to prevent uncoupling of the mechanism if the handle is rotated one revolution below the lowest torque value.

3.18.1 Accuracy calibration and sealing. Type III wrenches shall be correctly calibrated at the manufacturer's plant in accordance with the accuracy requirements of 3.12.1, after which the type III wrenches shall be protected against tampering by sealing the adjusting mechanism with a fusible alloy of low melting point.

3.18.2 Locking device. Type III wrenches shall have a positive locking device which shall hold the torque setting securely during the operation of the wrench.

3.18.3 Tee-handles. Type III, classes 1 and 3, wrenches of 2000 foot-pounds maximum capacity shall be provided with two tubular extension handles and an adapter for tee-handled operation. Means shall be provided to lock the extensions and adapter in place.

3.18.4 Range. Type III wrenches shall conform to table VII for the size specified (see 6.2). Wrenches of a torque capacity, or drive size other than those listed in table VII shall be as specified (see 6.2). Wrenches of a greater range meeting the other requirements of a specific size and class will be acceptable and shall be tested throughout the entire range.

TABLE VII. Type III, Micrometer adjustable, torque presetting, (feel impulse)

Torque capacity (min.)		Square drive	Increment graduations (max.)	Class				
				1	2	3	4	
Inch-pounds	Foot-pounds	Inch						
5-50	----------	1/4	1 in.-lbs.	x	x	x	x	
30-150	----------	1/4	2 in.-lbs.	x	x	x	x	
150-750	----------	3/8	5 in.-lbs.	x	x	x	x	
700-1600	----------	1/2	10 in.-lbs.	x	x	x	x	
1200-4800	----------	3/4	40 in.-lbs.	x	x	x	x	
----------	15-75	3/8	1/2 ft.-lbs.	x	x	x	x	
----------	30-150	1/2	1 ft.-lbs.	x	x	x	x	
	60-300	3/4	5 ft.-lbs.	x	x	x	x	
----------	100-500	3/4	5 ft.-lbs.	x	x	x	x	
----------	500-2000	1	10 ft.-lbs.	x		x		

X = Available in this class.

3.18.5 Type III, class 1, micrometer adjustable, torque presetting. Type III, class 1 wrenches shall indicate torque in a right-hand (clockwise) direction of operation and be capable of wrenching in a left-hand (counterclockwise) direction of operation to the maximum capacity of the wrench. Type III, class 1, wrenches shall have a plain head, and shall be similar to figure 14, except 500-2000 foot-pound capacity wrenches shall be similar to figure 15.

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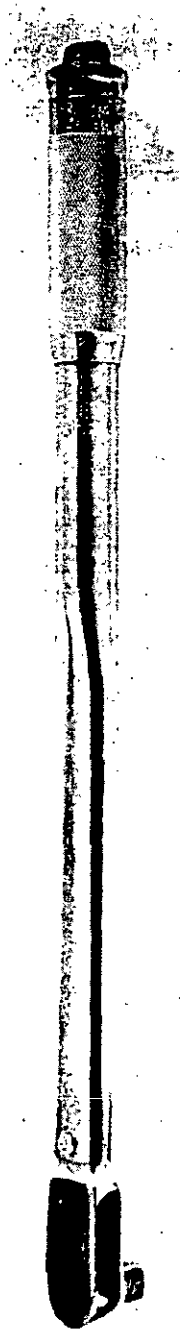


FIGURE 14. Type III, classes 1 and 3, torque wrench, micrometer adjustable, torque setting, plain head, nonratcheting.

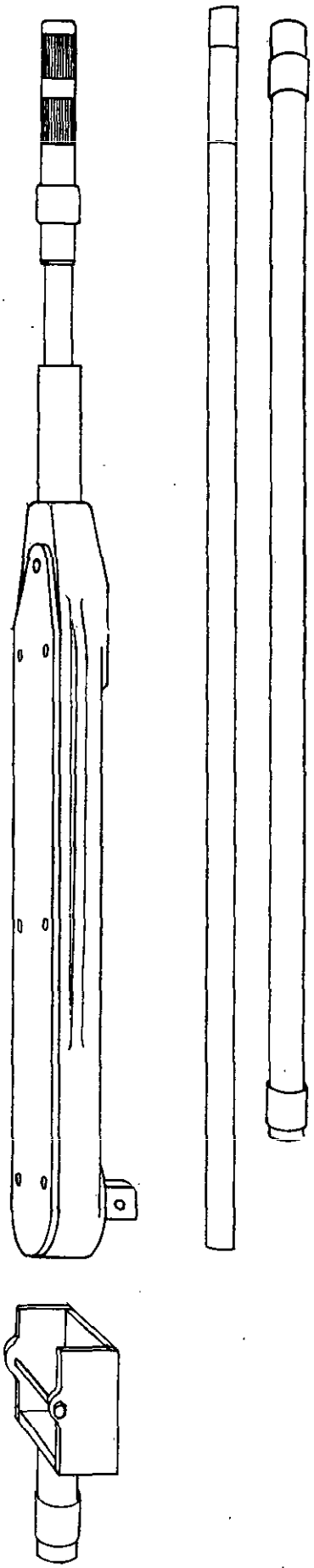


FIGURE 15. Type III, classes 1 and 3, torque wrench, tee-handle, (center drive) style, 500-2000 ft. - lb. capacity, micrometer adjustable, presetting, (feel impulse).

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3.18.6 Type III, class 2, ratchet, reversible head. Type III, class 2, wrenches shall conform to the requirements for class 1 wrenches, except that they shall contain a ratchet mechanism enclosed in the drive head. The ratcheting device shall meet the requirements of GGG-W-641. The reversing device, pawl and ratchet wheel shall be of steel suitably hardened for hard usage. Class 2 wrenches shall be similar to figure 16.

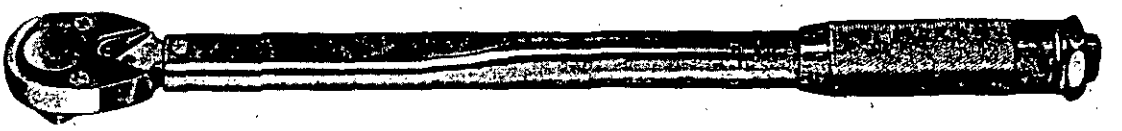


FIGURE 16. Type III, classes 2 and 4, torque wrench, micrometer adjustable, torque presetting, ratchet, reversible head.

3.18.7 Type III, class 3, micrometer adjustable, torque presetting. Type III, class 3, wrenches shall indicate torque in a right-hand (clockwise) and left-hand (counterclockwise) direction of operation to the maximum capacity of the wrench. Type III, class 3 wrenches shall have a plain head, and shall be similar to figure 14, except 500-2000 foot-pound capacity wrenches shall be similar to figure 15.

3.18.8 Type III, class 4, ratchet, reversible head. Type III, class 4, wrenches shall conform to the requirements for class 3 wrenches, except that they shall contain a ratchet mechanism enclosed in the drive head. The ratcheting device shall meet the requirements of GGG-W-641. The reversing device, pawl and ratchet wheel shall be of steel suitably hardened for hard usage. The class 4 wrenches shall be similar to figure 16.

3.19 Type IV, tee-handle, rigid case, ratcheting, preset. Type IV wrenches shall contain totally enclosed measuring elements mechanically linked with the drive square or tang, and shall have a smooth tee type handle. The wrenches shall operate by releasing completely at the preselected torque value instantly followed by automatic reset when rotated in the right-hand (clockwise) direction which shall occur no less than 6 times in one full revolution. An integral ratchet shall operate when the wrench is rotated in the left-hand (counterclockwise) direction. Type IV wrenches shall conform to the requirements of table VIII and shall be similar to figure 17. Wrenches of a torque capacity or drive size, other than those listed in table VIII, shall be, as specified (see 6.2).

TABLE VIII. Type IV, tee-handle, ratcheting, audible signal, torque value set and sealed

Size	Torque capacity		Square drive
	Inch-ounces	Inch-pounds	Inch
1	5-80	----------	1/4
2	----------	5-25	1/4
3	----------	25-60	1/4
4	----------	25-90	3/8

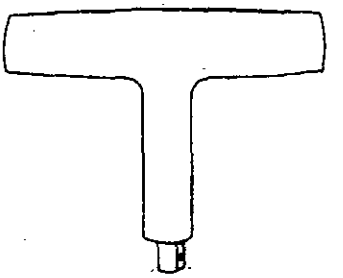


FIGURE 17. Type IV, torque wrench, tee-handle, rigid case, ratcheting, torque value set and sealed.

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3.19.1 Sealed torque setting. When the torque setting is specified (see 6.2), which shall be within the ranges specified in table VIII, the wrench shall be set to the desired torque in accordance with the accuracy requirements of 3.12.1 and sealed at the manufacturer's plant.

3.19.2 Sealing compound. Each type IV wrench shall be supplied with a sufficient quantity of low melting point fusible alloy which can be melted in boiling water to seal the adjustment accesses.

3.19.3 Sealing and marking. After the wrench is set or reset to the desired torque requirements, the compound shall be poured into the ends to seal the adjustment accesses. It shall then have the torque setting stamped or marked on the handle.

3.20 Instructions. Unless otherwise specified (see 6.2), the manufacturer shall furnish one copy of instructions with each wrench.

3.21 Workmanship. Workmanship shall be of the highest grade throughout and equal in every respect to good commercial practice.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except, as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Inspection of materials and components. In accordance with 4.1 above, the supplier is responsible for insuring that materials and components used were manufactured, tested, and inspected in accordance with the requirements of referenced subsidiary specifications and standards to the extent specified herein, or, if none, in accordance with this specification.

4.2 Sampling procedures. Sampling procedures shall be in accordance with MIL-STD-105. Data for sampling shall be as stated in table IX.

TABLE IX. Sampling data

Category	Sample unit	Inspection level	Acceptable quality level	AQL expressed in terms of	Reference
Visual examination	1 ea.	II	Major 2.5 Minor 4.0	Defects per hundred units	4.3.1
Dimensional examination	1 ea.	II	Major 2.5 Minor 4.0	Defects per hundred units	4.3.2
Testing (Except endurance)	1 ea.	S-2	2.5	Percent defective	4.4
Testing (endurance)	1 ea.	S-1	4.0	Percent defective	4.5

4.3 Examination.

4.3.1 Visual examination. Each sample unit shall be examined for any nonconformance in design, material, finish, coating, construction, workmanship, and marking. Defects are listed in table X.

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TABLE X. Classification of defects

Categories	Defects
Critical:	
1	None defined.
Major:	
101	Type, class, size and drive square not as specified.
102	Materials defective or not as required; evidence of cracks, deep pits, voids, heavy scale, deep tool marks, porosity, or foreign inclusions.
103	Scale illegible or not as specified; graduations not uniform, not adequately spaced, not permanent, or graduations not identified (numbered) as required; crystal (if applicable) scratched, not transparent, or crystal not unbreakable.
104	Pointer(s) bent, malformed, or not readily visible.
105	Male tang defective, not provided with a spring-loaded ball (or plunger), or ball (or plunger) not staked in place.
106	Type of signal mechanism (if applicable) utilized for producing the torque indication not as specified for the particular type.
107	Torque setting mechanism (if applicable) cannot be readily adjusted or setting (if required) cannot be locked.
108	Ratcheting mechanism defective (as applicable).
109	Cover not secure or cannot be readily removed for inspection and replacement of parts or not provided with an adapter handle (where applicable).
110	Torque setting not sealed or torque setting not legibly stamped on the handle (type IV, when required).
111	Marking, model number, manufacturer's name or trademark missing, illegible, incorrect, or not permanent.
112	Indicator not marked as specified.
113	Measuring element mechanism not suitably enclosed to prevent entrance of foreign matter (where applicable) (see 3.4).
Minor:	
201	Wrench not smoothly finished; evidence of fins, burrs, rough surfaces or sharp edges.

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4.3.2 Dimensional examination. Each sample unit shall be examined for any nonconformance with dimensional requirements.

4.4 Testing. Each sample unit shall be tested in accordance with 4.4.1 thru 4.5.3 and in addition, one sample unit from each sample shall be tested in accordance with 4.4.5, as applicable.

4.4.1 Overload. Types I, II, and III wrenches shall be subjected to an overload of 25 percent over rated capacity in both clockwise and counterclockwise directions, prior to conducting the accuracy test specified in 4.4.3.1, 4.4.3.2, or 4.4.3.3, as applicable.

4.4.2 Measurements. All torque accuracy measurements shall be made with either a dead weight testing device or with an elastic calibration testing device. A dead weight testing device is defined as a tester employing the use of a lever and weight coupled to produce torque. An elastic calibration testing device is defined as a tester utilizing the principle that stress is essentially proportional to strain. The accuracy of the tester shall be at least four times as accurate as the required wrench tolerance. For example, if the required wrench accuracy at 100 foot-pounds is ± 4 foot-pounds, the tester must be accurate to ± 1 foot-pound at a reading of 100 foot-pounds. The tester scale shall be such that interpolation between graduation lines will not be required to determine readings to closer than one-half of the increment between the graduation lines. The test load shall be applied smoothly at a rate not to exceed 15° per second.

4.4.3 Calibration accuracy test. The calibration accuracy test must be performed prior to any other test, except overload test 4.4.1, for the accuracy specified in 3.12.1.

4.4.3.1 Types I and II wrenches. Each sample wrench shall be subjected to a calibration accuracy test at the test loads given in table XI in a right-hand direction. The test shall be repeated in a left-hand direction. The wrench shall be flexed once to approximately one-half load in the direction being tested, removed from the tester and then the indicator dial or pointer shall be set on zero, if necessary, when the wrench is in a horizontal plane with the dial face up. The wrench shall be secured to the tester and a load applied to the center of the handgrip until the pointer of the wrench is aligned with the prescribed graduation line for the first test load position of table XI. The tester reading shall be recorded and the load increased to the next load position and this tester reading recorded. This procedure shall be continued until all readings are obtained. The load shall then be removed and the entire procedure duplicated two more times beginning with the zeroing of the wrench. The arithmetical average of the three tester readings at each test load position shall be recorded as the true readings of the wrench. These true readings must be within the tolerance specified in table I.

4.4.3.2 Types I and II wrenches with signal mechanism. Each sample wrench shall be subjected to the test of 4.4.3.1, except the force shall be applied to the center of the handgrip until the signal operates. The indicated torque shown by the tester at the instant of the signal shall be compared to the torque setting of the wrench. The test loads shall be applied to verify the accuracy at each test value.

4.4.3.3 Type III wrenches. Each sample wrench shall be tested in a right-hand direction and shall be set to the first test load specified in table XI. and secured to the tester. The load shall be increased until the instant the signal mechanism is actuated. At this instant the tester value shall be recorded. The test load shall be removed and the above procedure duplicated two more times. The arithmetical average of the three tester readings shall be recorded as the true readings of the wrench. The wrench shall then be adjusted to the next test load and the procedures repeated until all readings are obtained. These true readings must be within the limits specified in table I. The test shall be repeated in a left-hand direction for wrenches capable of indicating torque in both right-hand and left-hand directions. Wrenches capable of torquing in a right-hand (clockwise) direction only shall be tested for compliance with 3.18.5 for wrenching in a left-hand (counterclockwise) direction.

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TABLE XI. Test loads

Types	*Size	*Test loads for each size				
		Percentage of full capacity (nominal)				
		20	40	60	80	100
I and II	0-5	1	2	3	4	5
	0-10	2	4	6	8	10
	0-15	3	6	9	12	15
	0-16	4	6	10	12	16
	0-25	5	10	15	20	25
	0-30	6	12	18	24	30
	0-32	8	12	18	24	32
	0-50	10	20	30	40	50
	0-60	12	24	36	48	60
	0-75	15	30	45	60	75
	0-80	20	30	45	60	80
	0-100	20	40	60	80	100
	0-120	25	50	70	100	120
	0-150	30	60	90	120	150
	0-160	40	60	100	130	160
	0-175	35	75	105	140	175
	0-200	40	80	120	160	200
	0-250	50	100	150	200	250
	0-300	60	120	180	240	300
	0-350	70	140	210	280	350
	0-420	90	160	250	340	420
	0-425	90	160	250	340	425
	0-500	100	200	300	400	500
	0-600	120	240	360	480	600
	0-760	160	300	440	600	760
	0-1000	200	400	600	800	1000
	0-1200	240	500	700	1000	1200
	0-1800	360	720	1080	1440	1800
	0-2000	400	800	1200	1600	2000
	0-2400	500	1000	1450	1900	2400
	0-3600	750	1400	2100	2900	3600
	0-4200	840	1680	2520	3360	4200
	0-7200	1440	2880	4320	5760	7200
III**	5-50	10	20	30	40	50
	5-75	15	30	45	60	75
	15-75	15	30	45	60	75
	5-150	30	60	90	120	150
	10-150	30	60	90	120	150
	30-150	30	60	90	120	150
	60-300	60	120	180	240	300
	100-500	100	200	300	400	500
	100-750	150	300	450	600	750
	150-750	150	300	450	600	750
	700-1600	---	700	960	1280	1600
	200-2000	400	800	1200	1600	2000
	300-2000	400	800	1200	1600	2000
	500-2000	500	800	1200	1600	2000
	1200-4800	1200	1900	2900	3800	4800

*Inch-grams, inch-ounces, inch-pounds or foot-pounds as applicable.
**To be also tested at lowest calibrated scale reading and maximum range if greater.

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4.4.3.4 Type IV. Each sample wrench shall be tested at the value for which the wrench is set by rotating it one revolution and checking each of the torque values produced on a torque tester. The accuracy shall be as specified in 3.12.1. Wrenches not preset at the manufacturer's plant (see 6.2) shall be tested through one revolution for minimum capacity and through one revolution for maximum capacity.

4.4.4 Hardness test for drive squares. Each of the samples selected in accordance with 4.2 shall be subjected to a hardness test to determine conformance with the hardness value for drive squares, as specified in 3.7. Test shall be in accordance with Method 243.1 of Fed. Std. No. 151.

4.4.5 Nonmetallic grips. When wrenches having handgrips of plastic or material other than wood or metal are submitted, the handgrip of one wrench selected in accordance with 4.4 shall be submitted to the tests specified in 4.4.5.1 and 4.4.5.2. If the wrench fails the test, another wrench from the sample unit shall be tested. Failure of the second wrench to pass the test shall be cause for rejection.

4.4.5.1 Shatter test. Handgrip shall be placed on a solid foundation and a suitable guard to protect the operator of the test shall be placed around the handgrip. A weight of approximately 5 pounds shall be dropped directly upon the handgrip from a 3-foot height. Test shall be repeated. If handgrip is still intact after the two direct blows, it shall have passed the shatter test.

4.4.5.2 Flammability test. Handgrip shall be clamped in a vise and a flame from blowtorch, Bunsen burner, or equivalent shall be applied to one end for 30 seconds. If this does not ignite the material, flame shall be applied the second time for 30 seconds. If the second application does not ignite the handgrip, it shall be considered as having passed the test. If the material ignites and the rate of burn does not exceed 2 inches per minute, it shall be considered slow burning, thereby also passing the test.

4.5 Endurance tests.

4.5.1 Type I and II wrenches. Type I and II wrenches shall be stressed from a no load condition to a full load condition 5,000 times in a clockwise direction and 5,000 times in a counterclockwise direction and without recalibration, they shall be subjected to the test of 4.4.3. The wrenches shall then be stressed from a no load condition to 80 percent of a full load condition 45,000 times in a clockwise direction and 45,000 times in a counterclockwise direction. The signaling mechanism shall be in operation during the tests. The wrenches shall be stressed at a rate of not less than 12 nor more than 120 times a minute. At the conclusion of the test, the wrenches shall be again subjected to the test of 4.4.3 without recalibration. Minor maintenance on exterior of wrench without recalibration such as may be performed without the use of special tools, shall be permitted after each 1,000 cycles during the test. Ratcheting mechanisms need not comply with the requirements of this cycling test but shall be tested in accordance with GGG-W-641.

4.5.2 Type III wrenches. Type III wrenches indicating torque in a clockwise direction only shall be stressed from a no load condition to a full load condition 10,000 times in a clockwise direction. Wrenches indicating torque in both a clockwise and counterclockwise direction shall be stressed from a no load condition to a full load condition 5,000 times in a clockwise direction and 5,000 times in a counterclockwise direction and without recalibration shall be subjected to the test of 4.4.3. After minor recalibration, wrenches indicating torque in a clockwise direction only, shall be stressed from a no load condition to 80 percent of a full load condition 90,000 times in a clockwise direction. After minor recalibration, wrenches indicating torque in both clockwise and counterclockwise directions shall be stressed from a no load condition to 80 percent of a full load condition 45,000 times in a clockwise direction, and 45,000 times in a counterclockwise direction. The wrenches shall be stressed at the rate of not less than 12 nor more than 60 times a minute. At the conclusion of the test, the wrenches shall be recalibrated and then subjected to the test of 4.4.3. Minor maintenance, such as may be performed without the use of special tools shall be permitted. Ratcheting mechanisms need not comply with the requirements of this cycling test but shall be tested in accordance with GGG-W-641.

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4.5.3 Type IV wrenches. Type IV wrenches shall be stressed from a no load condition to the maximum rated load 50,000 times at a rate of not less than 12 nor more than 120 times per minute. At the conclusion of the test, the wrenches shall be subjected to the test specified in 4.4.3.4.

4.6 Inspection of preparation for delivery requirements. An inspection shall be made to determine that the preservation, packaging, packing, and marking comply with the requirements in section 5 of this specification.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging and packing. Unless otherwise specified (see 6.2), the wrenches shall be preserved, packaged, and packed in accordance with MIL-H-15424. The level of preservation and packaging shall be A or C, and the level of packing shall be A, B, or C, as specified (see 6.2).

5.2 Marking.

5.2.1 Civil agencies. In addition to markings required by the contract or order, the unit packages and shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.2.2 Military activities. In addition to markings required by the contract or order, the unit packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. Torque wrenches are used to apply or measure torque to bolts, nuts, and capscrews in mechanical assemblies. Torque wrenches should be selected to use the upper 80 percent of their capacity for more accurate results.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Type, class, style (see 1.2), and size of drive square required, (see applicable tables).
(c) If 1 percent or 2 percent wrench accuracy is required (see 3.12.1).
(d) Whether carrying case is not required, and color of enamel, if other than manufacturer's standard (see 3.15).
(e) If wrenches of a different capacity or drive size are required (see 3.16.2, 3.16.3, 3.16.4, 3.17.1, 3.17.2.3, 3.18.4 and 3.19).
(f) If center drive is required for 0-600 foot-pound type I, class 2, wrench (see 3.16.3).
(g) Torque value at which type IV wrenches should be preset and sealed (see 3.19.1).
(h) If instructions are not required (see 3.20).
(i) Levels of preservation, packaging, and packing required (see 5.1).
(j) Marking, if different (see 5.2).

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6.3 Supersession data. The classification of torque wrenches covered in this specification, corresponding to those covered in GGG-W-686B are as follows:

GGG-W-686B	GGG-W-00686C(GSA-FSS)
Type I. Deflecting beam.	Type I. Deflecting beam, direct reading, end drive.
Class 1. Indicator plate.	Class 1. Indicator plate, plain head.
Style A. Direct reading.	Style A. Without signal, without memory indicator.
Style B. Direct reading, with audible signal.	Style B. With (feel impulse) signal.
	Style C. Without signal, with memory indicator.
Class 2. Indicator dial.	Class 2. Indicator dial, plain head.
Style A. Direct reading.	Style A. Without signal, without memory indicator.
Style B. Direct reading, with audible signal.	Deleted.
	Style C. Without signal, with memory indicator.
	Class 3. Indicator plate, ratchet reversible head.
	Style A. Without signal, without memory indicator.
Type II. Rigid case with indicator dial.	Type II. Rigid case, with indicator dial.
	Class 1. End drive.
Style A. Direct reading.	Style A. Without signal, without memory indicator.
Style B. Presetting torque dial, direct reading, with audible signal.	Deleted.
	Style B. With visual signal, without memory indicator.
	Style C. Without signal, with memory indicator.
	Class 2. Tee-handle (center drive).
Style C. Direct reading and (flash signal).	Style A. Without signal, without memory indicator.
	Style B. With visual signal, without memory indicator.
	Style C. Without signal, with memory indicator.
Type III. Rigid case, micrometer style torque presetting, audible indicating.	Type III. Rigid case, micrometer style, torque presetting, (feel impulse).
Class 1. Plain head.	Class 1. Plain head (RH).
Class 2. Ratchet reversible head.	Class 2. Ratchet reversible head (RH).
	Class 3. Plain head (RH and LH).
	Class 4. Ratchet reversible head (RH and LH).
Type IV. Tee-handle, rigid case, ratcheting, audible signal.	
Class 1. Torque value set and sealed.	Type IV. Tee-handle, ratcheting, preset.
Class 2. Torque value setting adjustable.	Deleted.
Type V. Rigid case, preset torque, audible indicating.	Deleted.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

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